

INDUSTRIAL ENGINEERING

Industry :

The term “industry” as defined in Section 2(j) of the Industrial Disputes Act, 1947 is as follows :

“Industry” means any business, trade, undertaking, manufacture, or calling of employers and includes any calling, service, employment, handicraft or industrial occupation or avocation of workman”.

Business :

may be defined as “an activity, in which different persons exchange something of value whether goods or service for mutual gain of profit.

It may also be defined as “an enterprise engaged in production and distribution of goods for sale in market or rendering services for a price.”

Engineering :

Engineering involves the application of the principles of science and mathematics to solve real world problems and to innovate new products and processes across

The American Engineers' Council for Professional Development (ECPD), has defined “engineering” as:

The creative application of scientific principles to design or develop structures, machines, apparatus, or manufacturing processes, or works utilizing them singly or in combination; or to construct or operate the same with full cognizance of their design; or to forecast their behavior under specific operating conditions; all as respects an intended function, economics of operation and safety to life and property.

➤ The American Institute of Industrial Engineers (**AIIE**) has defined the **Industrial Engineering** as “Concerned with design, improvement and installation of integrated systems of people, materials, equipment and energy.”

➤ Industrial Engineering is going to play a pivotal role in increasing the productivity. It is the engineering approach to the detailed analysis of the **use and cost of the resources of an organization**. The **main resources** are men, money, materials, equipment and machinery.

Activities Of Industrial Engineering

- **Selection of processes and assembling methods**
- **Selection and design of tools and equipment**
- **Design of facilities including plant location layout of buildings, machines and equipments material handling system, raw materials and finished goods storage facilities.**
- **Design and improvement of planning and control system for production, inventory, quality and plant maintenance and distribution systems.**
- **Developing a cost control system such as budgetary control, cost analysis and standard costing.**

- **Development of time standard, costing and performance standards**
- **Development and installation of job evaluation system**
- **Installation of wage incentives schemes**
- **Design and installation of value engineering and analysis system**
- **Operation research including mathematical techniques and statistical analysis**
- **Performance evaluation**
- **Organization and methods**
- **Project feasibility studies**
- **Supplier selection and evaluation**

Objective of Industrial Engineering

- **To establish methods for improving the operations and controlling the production costs**
- **To develop programs for reducing those costs**

Technique of Industrial Engineering

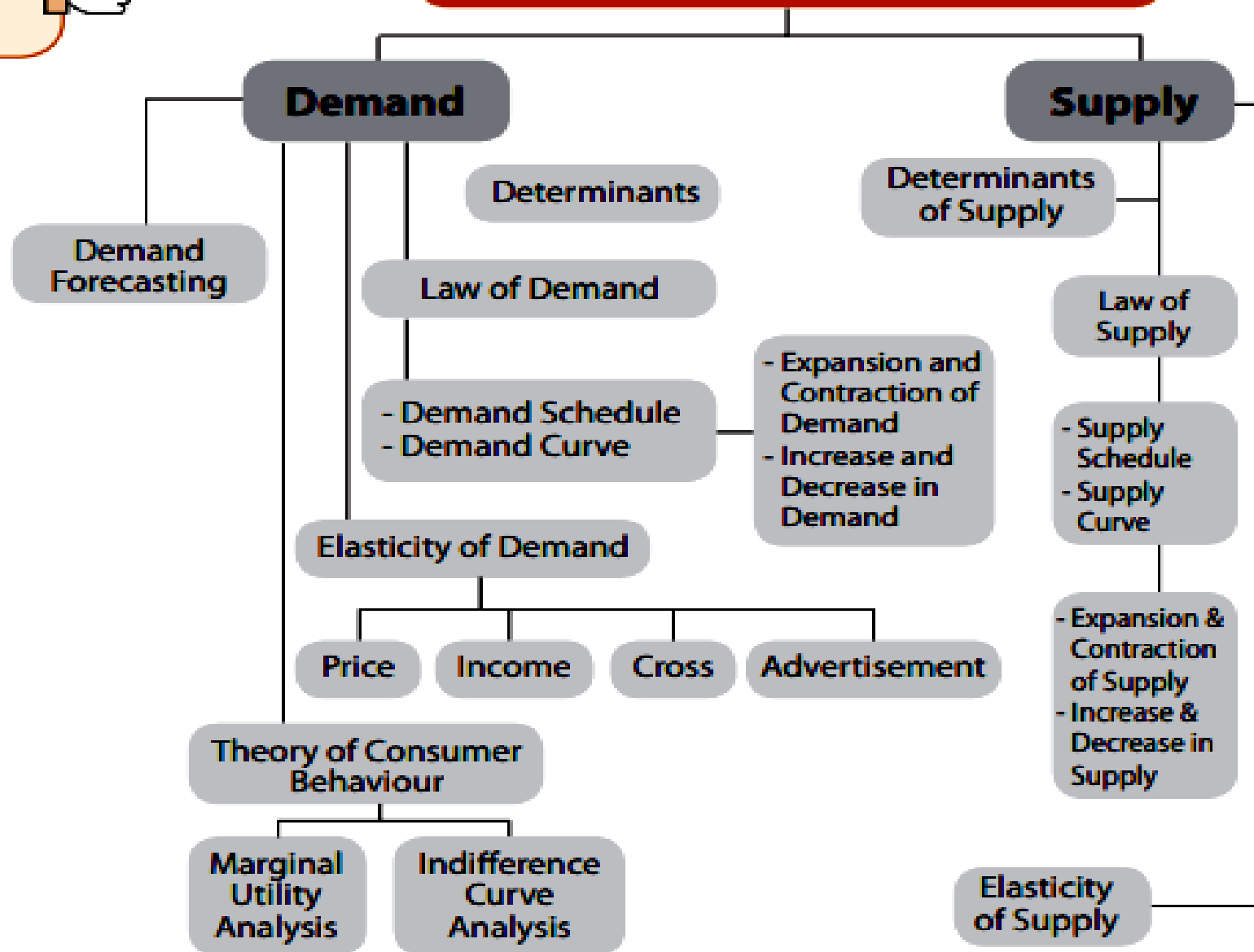
- **Method study**
- **Time study**
- **Motion study**
- **Financial and non-financial incentives**
- **Value analysis**
- **Production, planning and control**
- **Inventory control**
- **Job evaluation**
- **Material handling analysis**
- **Ergonomics(Human engineering)**

- **System analysis**
- **Operation research techniques**
- **Other techniques**

Applications of Industrial Engineering

- **In health services**
- **In government organizations**
- **In banking**
- **Others such as marketing, finance, purchasing, industrial relations etc**

THEORY OF DEMAND AND SUPPLY



Meaning Of Demand

The concept 'demand' refers to the quantity of a good or service that consumers are willing and able to purchase at various prices during a given period of time.

Factors affecting Demand

(i) Price of the commodity or goods: other things being equal, the demand for a commodity is inversely related to its price. It implies that a rise in the price of a commodity brings about a fall in the quantity purchased and vice-versa. This happens because of income and substitution effects.

- (ii) Price of related commodities:** Related commodities are of two types:
- (a) complementary goods** and
 - (b) competing goods or substitutes.**

Complementary goods are those goods which are **consumed together or simultaneously**. For example; tea and sugar, automobile and petrol and pen and ink. When two commodities are complements, a fall in the price of one (other things being equal) will cause the demand for the other to rise. For example, a fall in the price of petrol-driven cars would lead to a rise in the demand for petrol. Similarly, a fall in the price of fountain pens will cause a rise in the demand for ink. The reverse will be the case when the price of a complement rises. Thus, we find that, there is an inverse relation between the demand for a good and the price of its complement.

Two commodities are called **competing goods or substitutes** when they satisfy the same want and can be used with ease in place of one another.

For example, tea and coffee, ink pen and ball pen, are substitutes for each other and can be used in place of one another easily. When goods are substitutes, a fall in the price of one leads to a fall in the quantity demanded of its substitutes. For example, if the price of tea falls, people will try to substitute it for coffee and demand more of it and less of coffee i.e. the demand for tea will rise and that of coffee will fall. Therefore, there is direct or positive relation between the demand for a product and the price of its substitutes

(iii) Income of the consumer: Other things being equal, the demand for a commodity depends upon the money income of the consumer . In most cases, the larger the average money income of the consumer, the larger is the quantity demanded of a particular good.

The nature of relationship between income and quantity demanded depends upon the nature of consumer goods (**Normal goods**- Household furniture, clothing, automobiles, consumer durables and semi durables etc.

. **Essential consumer goods** such as food grains, fuel, cooking oil, necessary clothing etc **Durable Goods** such as a TV, car, house etc.).

(iv) Tastes and preferences of consumers: Consumers may perceive a product as obsolete and discard it before it is fully utilised and prefer another good which is currently in fashion. For example, there is greater demand for LCD/LED televisions and more and more people are discarding their ordinary television sets even though they could have used it for some more years

(v) Consumers' Expectations: Consumers' expectations regarding future prices, income, supply conditions etc. influence current demand. If the consumers expect **increase in future prices**, increase in income and shortages in supply, more quantities will be demanded. If they expect a fall in price, they will postpone their purchases of nonessential commodities and therefore, the current demand for them will fall.

Other factors: Apart from the above factors, the demand for a commodity depends upon the following factors:

(a) Size of population: Generally, larger the size of population of a country or a region, greater is the demand for commodities in general.

(b) Composition of population: If there are more old people in a region, the demand for spectacles, walking sticks, etc. will be high. Similarly, if the population consists of more of children, demand for toys, baby foods, toffees, etc. will be more.

(c) The level of National Income and its Distribution: Higher the national income, higher will be the demand for all normal goods and services. If the distribution of income is more equal, then the propensity to consume of the country as a whole will be relatively high indicating higher demand for goods.

(d) Consumer-credit facility and interest rates: Availability of credit facilities induces people to purchase more than what their current incomes permit them. Credit facilities mostly determine the demand for **durable goods** which are **expensive and require bulk payments** at the time of purchase. **Low rates of interest** encourage people to borrow and therefore demand will be more.

Apart from above, factors such as government policy in respect of taxes and subsidies, business conditions, wealth, socioeconomic class, group, level of education, marital status, weather conditions, salesmanship and advertisements, habits, customs and conventions also play an important role in influencing demand.